Data Analytics (DA)

Module -4

Python Assignment 1 - Data Structures - Strings & Tuples

Assignment Task Overview

Problem Statement:

This assignment focuses on understanding **Python string operations and tuple manipulation**, including concatenation, slicing, built-in methods, and tuple operations.

Tasks:

Strings (Concatenation, Slicing, and Other Methods) :

1. **String Concatenation:**

   Write a Python program that takes two strings, i.e., **string 1** "Hello" and **string 2** "*get name"* as input from the user, and concatenates them together. Display the concatenated string as the output.

   Sample Output:

   ```
   Enter your Name: Zara
   Hello Zara
   ```

   Now, concatenate string 3, "Welcome to Python programming," to the existing string and display the output string.

   Sample Output:

   ```
   Hello Zara, welcome to Python programming
   ```

2. **String Slicing and Indexing:**

   Write a Python program using the above concatenated string as input and perform the following tasks:

   a. Print the first character of the string.
   b. Print the last character of the string.
   c. Print the first 5 characters of the string.
   d. Print the last 11 characters of the string.
   e. Print the string in reverse.
   f. Use slicing and print the word "Python" from the existing string.

3. **String Methods:**

Write a Python program that takes a string, strM = "Python beginner tutorial," and performs the following tasks:

a. Convert the sentence to uppercase.
b. Convert the sentence to lowercase.
c. Use Capitalize and return the sentence to the original input form.
d. Count the total number of occurrences of character 't' in the string.
e. Replace all occurrences of "Python" with "Data Analytics" in the input string strM = "Python beginner tutorial."

Tuples (Creation, Modification and Access) :

Create the **1st tuple** with values -> **(10, 20, 30)** and the **2nd tuple** with values -> **(40, 50, 60)**:

a. Concatenate the two tuples and store it in "t_combine"
b. Repeat the elements of "t_combine" 3 times
c. Access the 3rd element from "t_combine"
d. Access the first three elements from "t_combine"
e. Access the last three elements from "t_combine"

Deliverables

Submit a project folder containing:

- Complete the assignment in Google Colab or Jupyter Notebook.
- For Google Colab, share the notebook link with "View" access. If using Jupyter, upload the notebook to Google Drive and ensure "View" access is enabled.
- Put both the Google Colab notebook in a Google Drive folder and share that folder with view access

Skills to be Evaluated and Mapping

Assignment Section	Core Skill	Sub-Skill	Proficiency
String Concatenation	Data Manipulation	Combining strings using `+` operator	Beginner
User Input Handling	Input/Output Operations	Using `input()` and displaying output	Beginner
String Indexing	Data Access	Accessing characters using index positions	Beginner
String Slicing	Data Extraction	Extracting substrings using slicing	Beginner
Reverse String	Data Transformation	Using slicing for reversing strings	Beginner
String Methods	Built-in Functions	`upper()`, `lower()`, `capitalize()`	Beginner
Character Counting	Data Analysis	Using `count()` method	Beginner
String Replacement	Data Modification	Using `replace()` method	Beginner
Tuple Creation	Data Structures	Defining and initializing tuples	Beginner
Tuple Concatenation	Data Manipulation	Combining tuples using `+` operator	Beginner
Tuple Repetition	Data Operations	Repeating tuple elements using `*`	Beginner
Tuple Indexing	Data Access	Accessing elements using index	Beginner
Tuple Slicing	Data Extraction	Extracting subsets of tuples	Beginner
Immutability Concept	Data Integrity	Understanding tuple immutability	Intermediate

DA Skill Taxonomy Structure

Skill / Component	Full Marks	Criteria for Full Marks	Partial Credit Description	Partial Marks Range
String Concatenation & Input Handling	2	Correct use of input() and string concatenation with proper output display	Minor syntax errors or incorrect concatenation logic	1
String Slicing & Indexing	3	Accurate use of indexing and slicing (first, last, substrings, reverse, extracting "Python")	Some slicing/indexing operations incorrect or missing	1 – 2
String Methods Implementation	2	Correct use of upper(), lower(), capitalize(), count(), and replace()	One or more methods incorrectly implemented or missing	1
Tuple Operations	3	Correct creation, concatenation, repetition, and accessing elements of tuples	Minor mistakes in tuple operations or incorrect indexing/slicing	1 – 2

DA Skill Taxonomy Structure

- **String Operations:** Perform concatenation, slicing, indexing, and string transformations
- **Input/Output Handling:** Accept user input and display formatted results
- **Built-in Functions:** Use string methods like upper(), lower(), capitalize(), count(), and replace()
- **Tuple Operations:** Create, concatenate, repeat, and access tuple elements
- **Data Handling Concepts:** Understand indexing, slicing, and immutability of tuples

Evaluation Guidelines

Open and review the submitted **.py script** to ensure:

- All string operations (concatenation, slicing, methods) are correctly implemented
- Tuple operations are properly performed and outputs are accurate
- Code is executable without errors
- Proper use of Python syntax, indentation, and structure

For each section:

- Match student work to the specific **skill/sub-skill area** in the taxonomy
- Assign full or partial marks according to the rubric

Marking Criteria:

- **Full marks:** Correct logic, accurate output, clean and well-structured code
- **Partial marks:** Minor syntax errors, incomplete implementation, or incorrect logic

Additional checks:

- Ensure outputs match the expected results
- Check clarity, readability, and proper commenting in code
- Comment on strengths and areas of improvement for each section

Maintain objectivity—assess the **correctness, clarity, and logical implementation** of Python programs.

Achievement Criteria & Passing Rules

- At least **partial (non-zero) marks** must be obtained in every assignment section
- Students must fully achieve at least two major skills:
  - **String Operations**
  - **Tuple Operations**
- All Python programs must be **executable and logically correct**
- Proper input/output handling should be demonstrated

Feedback Requirement:

- A **detailed scorecard and feedback** must be provided for every submission